



FIELDWORK REPORT [SEMESTER V]

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**SCHEDULED TRIBES
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ABSTRACT

Milletts have historically been central to the diet and agriculture of tribal communities in Attappadi. However, shifts in food preferences, policy interventions, and socioeconomic factors have led to a decline in millet cultivation and consumption. This study evaluates the current status of millet-based agriculture and dietary patterns, with a particular focus on understanding why the 'Millet Village' program failed to achieve its intended goals. Additionally, the research examines the food security and nutritional status of pregnant women and children (0-5 years), identifying critical gaps in existing interventions.

Fieldwork and focus group discussions provided insights into evolving food habits, revealing a growing dependence on government-supplied rice and processed foods, which lack the nutritional richness of traditional millet-based diets. To address these concerns, we collaborated with the community and institutions to explore strategies for millet revival. As part of this engagement, we participated in a Young Adult Camp with AADI to understand youth perspectives on dietary changes and worked with a Coimbatore-based entrepreneur to develop millet-based snacks suited to modern preferences.

This research highlights the broader implications of food policy, cultural preservation, and nutritional security. By combining traditional knowledge with modern dietary preferences, this study proposes a sustainable approach to improving nutrition while preserving cultural food heritage, offering a potential framework for dietary interventions in tribal communities.

INTRODUCTION

This report presents the findings of a comprehensive study conducted in Attappadi, a tribal block located in the Palakkad district of Kerala, from December 26, 2024, to January 4, 2025. The research examines traditional agricultural practices, changing dietary patterns, governmental interventions, and their collective impact on the health and well-being of tribal communities in the region.

Attappadi is a tribal development block situated on the eastern sloping plateau in the Western Ghats, in Mannarkad taluk of Palakkad district of Kerala, covering an area of about 745 sq. km. The population consists of both adivasis and non-adivasis, with the Adivasi communities comprising three main groups: the Kurumbas (categorised as a 'particularly vulnerable tribal group'), the Mudugas, and the Irulas. Across Attappadi, there are 197 hamlets known as 'oorus', inhabited by both adivasis and non-adivasis.

Through a qualitative and descriptive research approach, this study investigates the status of millet cultivation, the extent of food security, and the nutritional status of residents across multiple tribal settlements (Unnatis). Collecting data from government officials, community development workers, frontline health providers, and tribal community members, the research provides a holistic understanding of the region's socioeconomic dynamics.

The findings reveal significant shifts in agricultural practices and dietary patterns influenced by land alienation, changing employment opportunities, government interventions, and cultural assimilation. Traditional millet-based diets have largely been replaced by rice-based consumption, with implications for nutritional outcomes. The report also examines the effectiveness of various government initiatives, including the Millet Village program, community kitchens, and nutritional monitoring systems, in addressing these challenges.

OBJECTIVES

- To evaluate the current status of millet cultivation and consumption
- To look into the reasons why Millet Village was not a successful program
- To study the food security and nutritional status of pregnant women and children (0-5 yrs)

METHODOLOGY

1. Research Design

Our study follows a qualitative and descriptive research design to examine the status of millet cultivation, the extent of food security and nutritional status in Attappadi. The study was conducted over a ten-day- period from 26/12/2024 till 04/01/2025 and focused on understanding the interactions between traditional agricultural practices, changing dietary patterns, governmental interventions, and their collective impact on the tribal communities' health and well-being.

2. Study Setting

The research was conducted in Attappadi, a tribal block located in the Palakkad district of Kerala. The study encompassed multiple sites, including:

- Government offices (Krishi Bhavan, ICDS)
- Non-governmental organisations (Kudumbashree, AADI)
- Tribal settlements (Vattalacky Unnati, Sambarkod Unnati, Vechapathi Unnati)

The region consists of 93 hamlets containing 197 Unnatis (tribal settlements), with each settlement having distinct characteristics while sharing common challenges related to land rights, agricultural practices, and nutritional status.

3. Sampling Strategy

The study employed a combination of sampling methods to select participants:

Purposive Sampling

Purposive sampling was used to select key institutional informants who could provide rich, relevant information about the research topics. This approach ensured representation from:

1. Government officials: Agricultural officers, ICDS personnel
2. Community development workers: Kudumbashree staff, AADI representatives
3. Frontline workers: Anganwadi workers, health promoters

Availability Sampling

For tribal community members, a convenience and availability sampling approach was adopted. During visits to tribal hamlets (Vattalacky Unnati, Sambarkod Unnati, Vechapathi Hamlet), interviews were conducted with community members who were present and willing to participate at the time of the visit. This opportunistic approach allowed for spontaneous interactions that captured everyday perspectives rather than only those of designated community leaders.

This sampling approach allowed for triangulation of data from multiple perspectives, enhancing the validity of the findings by including both formally selected institutional representatives and community voices.

4. Data Collection Method

In-depth Interviews

Semi-structured in-depth interviews were conducted with key informants, including:

- Vishnu, Agricultural officer at Agali Krishi Bhavan
- Kudumbashree, Assistant Project Officer, and Joemon, Social Development Coordinator
- Tribal community members (including the son of a chieftain)
- Fr Lenin Antony, AADI Assistant Director
- Jayanthi C R, ICDS Officer in charge of Child Development
- Anganwadi workers

The study used open-ended questions to explore their experiences, opinions, and challenges. Allowed for interaction and collective meaning-making among participants.

Focus Group Discussions

FGDs were conducted separately with tribal women, youth and NGO workers to understand different perspectives on millet cultivation, food security, and government interventions. Open-ended questions were used to explore participants' experiences, opinions, and challenges, allowing them to express their perspectives in depth. This approach facilitated interaction

among participants, enabling collective interpretation and a richer understanding of the issues discussed.

Direct Observation

Participant observation was employed during visits to tribal settlements, community kitchens, and agricultural sites to understand the lived experiences and daily practices of the community.

Document Review

Secondary data, including program reports, nutritional status records, and intervention documents, were reviewed to complement primary data collection.

5. Ethical Considerations

The research adhered to ethical principles, including:

- Informed consent was obtained from all participants
- Cultural sensitivity was maintained throughout the research process
- Privacy and confidentiality of respondents were protected
- Research activities were conducted with minimal disruption to community activities

6. Limitations of Study

The research methodology had several limitations:

- The short duration (10 days) limited the depth of immersion possible in the community.
- Seasonal variations in agricultural activities could not be observed directly, as the harvest was already done during our field visits.
- The sample size, while diverse, represents only a fraction of the tribal population.
- We were unable to conduct a quantitative study as we are still awaiting official data.

VILLAGE / HAMLET VISITS

Vattalacky Unnati (Hamlet)

Vattalacky is a village (Ward No. 3) in Sholayoor Gram Panchayath of Attappadi Block. As far as farming is concerned, Vattalacky used to have more area under agriculture but has recently witnessed a decline in areas under cultivation. The fieldworkers conducted multiple interviews and discussions with the chieftain, former ST promoter and other people in the village to understand the land alienation, agricultural shift and nutritional status.

Interviews conducted in Vattalacky presented the ongoing challenges faced by the Adivasi community due to land alienation. According to Murugan, the son of the village chieftain, the Adivasi population has declined significantly over time, from approximately 200,000 to around 30,000. He added that while laws and policies were introduced to protect tribal land rights, their implementation has been ineffective. Climatic changes and soil erosion have further reduced the availability of agricultural land, forcing shifts in traditional farming and dietary habits.

Surya, a former ST promoter in the village, reported that the Millet Village programme was effectively implemented during her tenure. However, she noted that the Agricultural Department failed to maintain follow-up measures, leading to a decline in participation. As a result, fewer people engaged in millet farming over time.

The introduction of the Mahatma Gandhi National Rural Employment Guarantee Act (MGNREGA) in Attappadi, which guarantees 200 days of work in tribal areas, has had unintended consequences. The availability of regular wages has reduced interest in agricultural activities, as farming is seen as less secure. Additionally, the presence of community kitchens has influenced food culture. Many villagers now rely on these meals rather than preparing food at home, despite having adequate food supplies. Meat consumption remains low, with most families consuming it only once a month.

Most farmers in Vattalacky are older adults, with minimal involvement from younger generations. Multiple factors, including employment opportunities through government schemes, declining interest in traditional occupations, migration, etc, have influenced the shift away from farming.

Sambarkode Unnati (Hamlet)

Sambarkod village also comes under the Sholayoor Gram Panchayath of Attappadi. Sambarkode has relatively more land under cultivation than Vattalacky and is not much affected by people shifting away from agriculture. The sowing period typically occurs during the rainy season (June-July), with harvesting in November-December. The early months of the year are dedicated to clearing land for farming.

In Sambarkod, increased natural reforestation has led to agricultural lands being classified as reserved forest areas. Murugan, a resident of the village, explained that farming practices are highly dependent on rainfall. Villagers sow seeds whenever it rains, following traditional patterns. However, there is a noticeable decline in the consumption of millets like ragi and chama.

Despite the declining preference for millets, government schemes such as the Millet Village project and 'Nammth Vellame' have encouraged continued cultivation. Farmers who participate in millet production benefit from electric fencing under government schemes, which helps protect crops from wild animals, although it does not completely prevent damage. After consumption and seed storage, surplus millet produce is sold in the market at minimum support prices.

The local Anganwadi provides millet-based meals (ragi) twice a week for children and ensures other nutrient-rich foods are available throughout the week. The introduction of the digital ODK app has improved the monitoring of nutritional status among children and pregnant women. Supplies for pregnant women and adolescent girls are distributed as part of government nutrition schemes.

Vechappathi Unnati (Hamlet)

During the field visit to Vechapathi Ooru/Unnati(hamlet), interviews were conducted with tribal women to understand their agricultural practices, food consumption patterns, and socio-economic conditions. The following observations and statements are based on their accounts and reflect their lived experiences.

Agriculture was once a predominant activity in the Ooru/Unnati(hamlet), with the women recalling a time when extensive cultivation took place. While wildlife attacks on crops remain a persistent challenge, they noted that pigs and smaller animals pose a greater threat than elephants. The impact of land fragmentation has also been significant. In earlier times, the

community had access to larger agricultural landholdings, which helped them sustain losses due to wildlife attacks. However, with the reduction in land size, even minor crop damage now leads to substantial losses, making agriculture increasingly difficult.

Traditional millet cultivation has nearly disappeared from the region. The women stated that their ancestors had cleared forest lands for farming, and in the past, agricultural land was communally owned. Today, land ownership has shifted to a single-ownership model, which has altered farming practices. Additionally, while their crops typically require minimal water, the seeds (of vegetables and medicinal) provided by external agencies, such as Vivekananda Hospital, have not been suitable for their land, as they require substantial water input, which is unavailable in the Ooru/Unnati (hamlet).

Rice has replaced millets as the primary staple food, even though it is not locally cultivated. The women procure rice through the Public Distribution System (PDS) and acknowledge that, despite their remote location, the system functions efficiently. They receive 30 kg of rice and 4 kg of wheat per month through the PDS, which has significantly improved food security. However, they noted that certain essential items such as kerosene, sugar, and jaggery are not included in the ration. However, nutritious food, along with sugar and jaggery, is provided to children through the Anganwadi system.

The women emphasised that their farming practices remain organic, with no use of chemical fertilisers or pesticides. However, shifts in staple food preferences have occurred over generations. Earlier, the community primarily consumed 'Chamma' rice (little millet), but they have now transitioned to 'Matta' rice. Despite these changes, younger generations continue to engage in agriculture, although on a smaller scale.

The women emphasised the health benefits of 'Ragi Kali' (ragi balls), which they said gave strength and stamina. However, many children now stay in hostels for education, where they are served Matta rice instead of millet. Hostel authorities reportedly find it difficult to prepare millet-based meals for large groups, leading to the gradual disappearance of these foods from children's diets.

Dietary habits have also evolved with changing market access. Meat has traditionally been a part of their diet, and in recent years, fish consumption has increased due to improved access to markets. While the women did not report direct conflicts with nearby tourist resorts, they claimed that a significant portion of their agricultural land had been encroached upon by outsiders.

The community remains dependent on rain-fed agriculture, making them vulnerable to climatic variations. This year, agricultural output was low due to insufficient rainfall, and the absence of irrigation facilities further exacerbated the issue. Despite these challenges, some crops continue to grow naturally without intervention, such as 'Cheera' (leafy greens), which the women collect and consume.

Rain-fed agriculture remains their only option, making them highly vulnerable to fluctuations in rainfall. This year, poor rainfall has led to a sharp decline in yield. Without irrigation facilities, they have no alternative water sources for cultivation.

The women reflected on past periods of food scarcity when they had to survive by boiling and eating wild leaves. In times of extreme need, they borrowed money from Tamil settlements to purchase essentials. They acknowledged that food security has improved with the PDS but emphasised that their situation remains precarious.

Healthcare services in the hamlet are accessible to some extent. The women reported that a pregnant woman in their community is receiving proper medical attention, including regular check-ups. They also mentioned the active presence of ASHA workers and health promoters, who play a crucial role in maternal and child healthcare. The community kitchen, which had been non-functional for some time, has recently been restocked and made operational again.

The women described their dietary patterns, noting that they typically consume two meals a day—one in the morning before going to work and another in the evening upon returning home. Their agricultural activities continue to follow the Malayalam calendar, aligning with traditional farming cycles.

Finally, they acknowledged the impact of government initiatives, particularly crediting A.K. Balan, former Minister of Electricity and SC/ST Development (2006–2011), for implementing projects that improved housing, road connectivity, and electricity access in the hamlet.

INSTITUTION VISITS

Krishi Bhavan, Agali Panchayath

An institutional visit was conducted to the Agali Krishi Bhavan (Agricultural Office) in Attappadi. The interview with Vishnu, an official at the office, provided valuable insights into agricultural governance, traditional practices, and challenges faced by tribal farmers in the region. Attappadi comprises 93 hamlets containing 197 Unnatis (tribal settlements), representing a significant administrative area for agricultural governance.

Agricultural Governance and Initiatives

Since 2021, agricultural governance in Attappadi has been brought under the Rebuild Kerala Project, marking a shift in administrative approach. The first systematic attempt to promote millets in the region began with the Rashtriya Krishi Vikas Yojana (RKVY), which implemented organic farming practices throughout Attappadi. Under this scheme, farmers were provided with Green Certificates to authenticate their organic farming status.

The Tribal Extension Office handles land-related matters, particularly addressing encroachment issues and land alienation affecting tribal communities. This office plays a crucial role in the redistribution of lands that have been encroached upon by settlers, highlighting ongoing land rights challenges in the area.

Traditional Agricultural Practices

The dominant traditional agricultural system in Attappadi is known as 'Panchakrishi', a mixed-cropping approach where five different crops are cultivated simultaneously. This indigenous practice typically includes two varieties of millet, mustard, spinach, and a pulse variety. This agricultural approach represents the most traditional form of farming in the region and reflects the ecological knowledge of the tribal communities.

Challenges in Agricultural Production

Human-wildlife conflict stands as a major challenge for farmers in Attappadi, resulting in significant reductions in agricultural yield. To address this issue, RKVY has implemented an electrical fencing project covering 997 hectares of agricultural land, with additional fencing schemes undertaken by the local Panchayat.

Despite the establishment of the Attappadi Farmer Produce Company (ATFAM) to provide market access for tribal farmers, Vishnu reported that it has underperformed in creating

effective market linkages. Additionally, delays in fund proposals for the Millet Village project have hampered its implementation, though the allocated funds are eventually disbursed in full.

Socioeconomic Impacts on Agriculture

A significant finding from the interview was the impact of the Mahatma Gandhi National Rural Employment Guarantee Act (MGNREGA) on agricultural practices. In Attappadi, MGNREGA offers 200 days of employment compared to the standard 100 days in other regions. Vishnu observed that many tribal people have shifted from farming to daily wage labour under MGNREGA due to the guaranteed daily income it provides, as opposed to the uncertainty and risks associated with agricultural production.

Nutrition and Social Welfare

The reintroduction of millet into the tribal diet has reportedly contributed to a reduction in infant mortality rates, suggesting important nutritional benefits from traditional food systems. However, Vishnu noted that the mid-day meal scheme in schools does not include millet, representing a missed opportunity for nutritional intervention among younger generations.

The interview also revealed that despite Attappadi being designated a liquor-prohibited area, illicit brewing of local alcohol (moonshine) has increased, attributed to the lack of alternative recreational activities. Additionally, Vishnu mentioned that consumption of Vettila (betel leaf) has been associated with increased abortion rates, highlighting complex socio-cultural factors affecting maternal and child health.

Kudumbashree

Kudumbasree was initiated in 2013 as a project specific to Attappadi. Its primary aim is to support local communities through initiatives that improve food security, promote traditional agricultural practices, and empower women in agriculture.

At the core of Kudumbasree's efforts is the establishment of community kitchens. Originally set up in 145 Unnatis, these kitchens now operate in fewer than 35 villages. The kitchens are supported by a dual funding model, with 60 per cent of the financial resources coming from the Women and Child Department and the remaining 40 per cent from the ST Department. These facilities serve millet-based meals for breakfast and dinner and provide rice-based meals for lunch. The consistent availability of nutritious meals is designed to mitigate food insecurity

among vulnerable populations, particularly in a region where dietary patterns have shifted over time.

In addition to the community kitchens, Kudumbasree implements the Mahila Kisan Shashakthikaran Pariyojana (MKSP), a component of the National Rural Livelihood Mission. Through this initiative, the organisation supports women in agriculture by forming Joint Liability Groups within Ayalkkoottam. These groups are provided with financial loans, which aim to promote agricultural practices and enhance the economic stability of women, who play a significant role in the local farming community. The emphasis on women's empowerment through MKSP is a strategic measure to ensure that traditional agricultural knowledge and practices are sustained even as modern influences alter the local food patterns.

Traditional practices are also given importance through Kudumbasree's promotion of communal farming events. One such event is 'Kambalam', during which community members work collectively on agricultural tasks. This is with the intention of sustaining traditional practices. The organisation also claims a role in reducing infant mortality rates by strengthening nutritional monitoring and support systems for mothers and children.

The operational effectiveness of Kudumbasree is further enhanced by its robust nutritional monitoring system. The organisation uses a digital ODK application to track the nutritional status of children aged 0 to 5, with data updates made at least four times a week. This systematic approach allows for timely interventions and adjustments to nutritional programs, ensuring that children receive the necessary nutrients for healthy growth.

Interviews conducted at the Kudumbasree office provide additional insight into the functioning of the project. An Assistant Project Officer highlighted the operational management of the community kitchens and stressed the challenges associated with cultural practices that delay the disclosure of pregnancy during the critical early months. The interviewee states that the major issue that they face is the cultural practices of a few Adivasis where they do not reveal their pregnancy for the first 3 months. These are months when primary intake of nutrient intake is necessary for pregnant women as they are prone to conditions like anaemia. Furthermore, Jomon, a Social Development Coordinator, discussed the broader socio-cultural impacts observed in the region. He explained that the influx of settlers had forced Adivasis deeper into the forest, leading to reduced agricultural holdings and a shift from traditional, extensive agriculture to smaller, more fragmented plots. This transition, coupled with the introduction of

rice through the Public Distribution System and other socio-cultural factors, has gradually changed the community's dietary patterns.

One of the critical health issues is the low average haemoglobin level among Adivasis. It is claimed that the consumption of beetle leaves may contribute to this deficiency, although this claim has not been scientifically proven. In addition to low haemoglobin levels, there is a reported increase in abortion rates. In 2024, there were 119 reported abortions, and medically suggested terminations—particularly those due to detected fetal anomalies—have also contributed to these statistics. Jomon also noted concerns such as low initial weight among pregnant women and insufficient weight gain during pregnancy despite nutritional interventions (it is ideal for a pregnant woman to gain 12 kgs towards the final trimester, but it was found that women in Attappadi only gained 5-7 kgs towards the final trimester), and related health issues like low haemoglobin levels and instances of sickle-cell anaemia. As the Kudumbasree report identifies, pregnant women constitute around 200 in Attappadi Adivasi every week, and their average weight at the initial stage of pregnancy is found to be around 30-32 kg. As per the standards kept by Kudumbasree, a pregnant should weigh at least 45 kg over the trimesters. Ideally, pregnant women should gain at least 12 kg towards the end of the third trimester, but even after nutrients are supplied, the scale of weight gained is still low. Among the average of 10 new entries of pregnant women every week, 2-3 of them tend to show traits of sickle-cell anaemia, where, in most cases, at least one tested positive for sickle-cell anaemia.

Children moving away from their homes to hostels for educational purposes consume a different dietary food pattern, with rice being the major component. This has led to a change in their consumption preferences. In most of the govt hostels, the menu is decided by government officials without considering their existing dietary patterns.

While food is available and affordable, the quality of nutrients in the diet remains questionable. Children identified as malnourished are classified into two categories: Severe Acute Malnutrition (SAM) and Moderate Acute Malnutrition (MAM). For those children suffering from SAM, the protocol involves admission to a Nutrition Rehabilitation Centre (NRC) for a period of two to three weeks, with a parent required to stay as a bystander. This requirement poses challenges for many families, especially when the mother is the sole breadwinner, as it disrupts their ability to work and support the household.

To address these issues, the nutritional status of children is closely monitored by a coordinated effort among four departments: ICDS, Kudumbasree, the Health Department, and ITDP. These

departments meet regularly to evaluate, monitor, and update the nutritional status of children, ensuring that interventions are timely and appropriate.

ICDS

Agriculture in Attappadi remains closely tied to the forest, but financial constraints have historically hindered the construction of protective fencing. This has led to frequent land encroachments by external groups, including organised land mafias. A key challenge is the lack of documented land ownership among the tribal population, making it difficult for them to claim legal entitlements. Without proper documentation, many families struggle to access government schemes meant to protect their rights.

The establishment of Anganwadi centres follows a need-based approach rather than population density. Even in remote areas with only a few households and children, Anganwadis are set up to ensure access to basic services. However, maintaining these centres and ensuring regular service delivery remains a challenge, particularly in isolated settlements.

Traditional cultural beliefs continue to influence health outcomes. Many families adhere to the practice of concealing pregnancies until after the first trimester, which can delay access to maternal healthcare. In cases where doctors recommend termination due to severe fetal abnormalities, parents often refuse, sometimes resulting in stillbirths. Another prevalent habit is the chewing and swallowing of betel leaves, which negatively impacts intestinal health and nutrient absorption, exacerbating malnutrition. Young children tend to mimic this habit, sometimes substituting betel leaves with tobacco or cigarettes, further affecting their health.

Cultural shifts are evident in various aspects of tribal life. Historically, festivals such as Ayyappan Vilakku and Maha Shivaratri provided opportunities for trade and interaction between tribal and non-tribal populations. However, increased cultural assimilation has led to a decline in these exchanges. Traditional attire is now mostly worn only by those above the age of 60, and inter-caste and inter-tribe marriages have contributed to a blending of physical features, making it harder to distinguish conventional tribal characteristics.

Education has also influenced lifestyle changes, particularly in dietary habits. Many children leave their settlements to attend schools and hostels, where they adopt mainstream food habits, particularly a rice-based diet. Upon returning home, they revert to consuming traditional crops such as millet. The Agricultural Department's 'Millet Village' initiative, active for the past two to three years, has had partial success but remains unevenly implemented. Meanwhile, the

availability of rice through the Public Distribution System (PDS) has made it the dominant staple, while traditional farming methods like "Karra Naellu Krishi" persist but contribute to only about 5% of the local food supply.

Despite the implementation of various nutritional programs, including maternal nutrition kits and growth monitoring, malnutrition remains a significant issue. Severe Acute Malnutrition (SAM) and Moderate Acute Malnutrition (MAM) cases currently stand at 2.9%. Although this percentage has slightly decreased, Attappadi's ranking in national nutrition indices has dropped from 91 to 112. The effectiveness of interventions for SAM and MAM cases is hindered by inadequate medical facilities and limited access to proper nutrition. Nutrition Rehabilitation Centers (NRCs) exist in three panchayats, but only two are fully operational. Previously, mothers staying at NRCs with their children received daily wages (around ₹200) as an incentive. However, this financial support was discontinued, leading to lower participation. Limited employment opportunities have resulted in parents relying on inexpensive snacks like biscuits to satiate their children's hunger, which negatively impacts nutritional rehabilitation efforts.

Government schemes such as Janani Janma Raksha and Pradhan Mantri Matru Vandana Yojana (PMMVY) provide financial assistance to pregnant women—₹5,000 for the first child and

₹6,000 for a second girl child. For Scheduled Tribes (STs), an additional ₹2,000 per month is given during the first 1,000 days post-conception. While these incentives are meant to support maternal and child health, concerns have emerged regarding their potential misuse. Some families, aware that stillbirths qualify for compensation of ₹1 lakh, may choose to proceed with high-risk pregnancies despite medical warnings, raising ethical concerns about the unintended consequences of such financial assistance.

Unlike conventional marriage traditions, tribal communities in Attappadi follow distinct social customs, particularly the 'Cheeru' function and funeral rites. 'Cheeru' marks the recognition of puberty and allows for flexible family structures, where individuals may form new relationships even after having children. While this practice provides social fluidity, it also raises concerns about the involvement of minors in relationships and the psychological impact on children whose parents may separate or form new partnerships.

Addressing malnutrition and overall health requires a holistic approach that considers not only physical but also mental and social well-being. External factors like climate change and settler migration have influenced socioeconomic transformations. While basic amenities such as water

and electricity are largely available, land disputes remain unresolved due to the absence of formal land ownership records. Tribal land was traditionally inalienable, but external groups have found ways to bypass these restrictions by financing cultivation and claiming a share of the yield. The introduction of fencing by settlers has further eroded tribal land ownership, as traditional tribal customs did not include defined territorial boundaries.

Healthcare access also presents challenges. Public hospitals in Attappadi prioritise natural deliveries, while private hospitals, which receive government funding for treating ST patients, often opt for cesarean sections due to liability concerns. A significant demographic issue is the stagnation of the ST population between 2010 and 2024, largely due to high mortality rates among men under 50. This mortality is often unrecorded in official statistics but is primarily linked to the increasing consumption of locally brewed and illicitly sourced alcohol. Over time, these beverages have become more potent, contributing to severe health deterioration and premature deaths.

Additionally, abortion rates in the region are often linked to recorded infant mortality statistics, raising concerns about possible underreporting or misclassification of reproductive health outcomes. These issues highlight the need for a more integrated approach to healthcare, land rights, and social welfare programs in Attappadi.

Millet Village

As part of our institutional visits, we attempted to engage with officials at the Millet Village office to gain insights into the program's implementation and challenges. However, the officials present were uncooperative and declined to provide any substantial information, instead directing us to the Krishi Bhavan. During subsequent discussions with other officials, we learned that the government had officially shut down the Millet Village program. Additionally, it was pointed out that the salaries of the officials previously associated with the initiative remained in arrears. This unresolved financial issue was suggested as a possible reason for their reluctance to engage in discussions or provide clarity on the program's status.

COMMUNITY ENGAGEMENT

Young Adult Camp with AADI

To engage the younger generation in the revival of millet consumption, we collaborated with AADI to conduct a **Young Adult Camp** focusing on skill enhancement and cultural values. The camp provided a platform for discussions on millet-based diets, emphasising their nutritional and historical significance within tribal communities. We also explored the evolving food preferences of young adults, identifying key factors influencing their dietary choices. Understanding these shifts was crucial in developing strategies that could make millet consumption more appealing and relevant to contemporary tastes and preferences while retaining its traditional significance.

PERSONAL REFLECTION

While requesting data from official sources, we faced significant difficulties as many were unwilling to share information. This made us realize how sensitive tribal health data is and how it has often been misused as a political tool. We chose not to rely on primary data because we didn't want to reduce tribal populations to mere numbers in our study. Every year, numerous independent research projects collect data on them, often without considering the repetitive nature of this process. Instead, we focused on qualitative data, using secondary sources for quantitative insights and complementing them with our qualitative findings (though we are still awaiting some secondary data).

A major discussion surrounding tribal health in Attappadi is the need to revive millet consumption. However, we must acknowledge that tribal communities are increasingly integrating with the mainland population, and their food habits are evolving. Expecting tribal children to avoid biscuits and chips while accepting these foods as normal for non-tribal children reflects a double standard.

Another key realization was that malnutrition can persist despite food security, as seen in Attappadi. While food is available, affordable, and accessible, malnutrition and nutrition-related issues still exist due to socio-economic factors such as land alienation, cultural practices, education-related migration etc.

One ethical dilemma we faced was the challenge of portraying tribal communities without either alienating them or over-glorifying their way of life. Often, narratives about tribals either romanticize their traditions or treat them as entirely separate from the mainstream, which can reinforce the idea of them as "the other." We wanted to avoid both extremes and instead highlight their evolving realities in a balanced manner.

Another thought-provoking moment came during a conversation with a Krishi Bhavan officer, who was tribal himself. He raised an important question: *"Do we continue promoting and respecting cultural practices if they harm health?"* He was referring to the practice of chewing and swallowing betel leaves, which is often defended as a cultural tradition but has known health risks. This made me reflect on the fine line between cultural preservation and necessary public health interventions—when does respect for tradition turn into the endorsement of harmful practices?

The hamlets we chose for our study varied in their geographical settings, which helped us observe differences and similarities influenced by location. Vattalakky was near the main road, Vachapathi was more remote, and Sambharkoodu fell somewhere in between. We also noticed that state policies significantly impact tribal communities, especially in hamlets along the Kerala-Tamil Nadu border, where conditions changed noticeably when crossing state lines.

Another important debate is whether promoting millets through commercialization will truly benefit tribal populations. While millet revival efforts are often framed as a way to support tribal communities, we need to critically examine who ultimately benefits from these initiatives. Additionally, there is the question of whether tribal youth are willing to continue their ancestral cultivation practices. Many prefer blue- or white-collar jobs that offer financial stability, and we cannot expect them to remain in agriculture simply because they belong to a tribal community.

This research made me realize that addressing tribal health and nutrition is not just about reviving traditional diets but also about considering socio-economic aspirations, state policies, and the ethics of research itself.

APPENDIX

